Biology Chapters B1–B5 — Cambridge IGCSE Biology Study Guide — Cheat Sheet

Key Terms

Organism	A living thing, such as a human, a platypus, a bacterium or a mango tree
Cell	The smallest unit from which all organisms are made
Unicellular	Made of just a single cell; bacteria and yeast are unicellular
Multicellular	Made of many cells; humans and plants are multicellular
Movement	An action by an organism or part of an organism causing a change of position or place
Respiration	The chemical reactions in cells that break down glucose and other nutrient molecules to release energy for the cell to use
Sensitivity	The ability to detect changes in the internal or external environment and respond to them
Growth	A permanent increase in size and dry mass, by the growth of cells and the addition of new cells
Reproduction	The processes that make new organisms of the same species
Excretion	The removal from the body of the waste products of metabolism (unwanted or toxic substances) and of substances in excess of requirements
Nutrition	Taking in substances from the environment and using them for energy, growth and development
Metabolism	All the chemical reactions that take place in an organism's cells, including respiration
Metabolic reactions	Chemical reactions that take place in living organisms
Light microscope	A microscope that shines light through the specimen and uses glass lenses to magnify and focus the image; a very good one magnifies about ×1500
Electron microscope	A microscope that can magnify far more than a light microscope; needed to see ribosomes, and to see mitochondria clearly
Photomicrograph	A photograph taken using a light microscope
Electron micrograph	A picture taken using an electron microscope, such as Figure B1.10
Cell membrane	A very thin layer of protein and fat surrounding the cytoplasm of every cell; it controls what enters and leaves the cell
Partially permeable	Allows some molecules and ions to pass through, but not others
Cell wall	A tough layer outside the cell membrane; found in the cells of plants, fungi and bacteria
Cellulose	A polysaccharide that forms criss-crossing fibres; plant cell walls are made mainly of cellulose
Fully permeable	Allows all molecules and ions to pass through it
Cytoplasm	The jelly-like material that fills a cell; it is nearly all water, with many substances (especially proteins) dissolved in it
Vacuole	A fluid-filled space inside a cell, separated from the cytoplasm by a membrane
Cell sap	The fluid that fills the large vacuoles in plant cells; a solution of sugars and other substances
Vesicle	A very small vacuole
Nucleus	A structure containing DNA in the form of chromosomes (plural: nuclei)
Chromosome	A length of DNA, found in the nucleus of a cell; it contains genetic information in the form of many different genes
DNA	A molecule that contains genetic information, in the form of genes, that controls the proteins that are made in the cell
Chloroplasts	Small structures found inside some plant cells, inside which photosynthesis takes place
Chlorophyll	The green pigment inside chloroplasts that absorbs energy from sunlight for photosynthesis
Starch grains	Structures that consist of thousands of starch molecules, stored in some plant cells
Ribosome	A very small structure in a cell that uses information on DNA to make protein molecules
Mitochondrion	A small structure in a cell where aerobic respiration releases energy from glucose (plural: mitochondria)
Aerobic respiration	Chemical reactions in mitochondria that use oxygen to break down glucose and other nutrient molecules to release energy for the cell to use
Bacteria	Unicellular organisms whose cells do not contain a nucleus (singular: bacterium)
Prokaryotic cells	Cells with no nucleus; bacteria have prokaryotic cells
Circular DNA	The single large circle of DNA found in a bacterial cell instead of chromosomes in a nucleus; sometimes called a bacterial chromosome
Plasmids	Small, circular molecules of DNA found in many prokaryotic cells in addition to the main, much larger circle of DNA
Peptidoglycan	The substance that bacterial cell walls are made of, instead of cellulose
Specialised cell	A cell whose structure is modified to help it carry out a particular function effectively
Tissue	A group of similar cells that work together to perform a particular function
Epidermis	The outer layer of tissue on a plant; also the outer layer of an animal's skin
Organ	A structure made up of a group of tissues, working together to perform specific functions
Organ system	A group of organs with related functions, working together to perform body functions
Digestion	The breakdown of food
Magnification	How many times larger an image is than the actual object
Micrometre (μm)	A unit of length equal to one thousandth of a millimetre; 1 mm = 1000 μm and 1 μm = 1×10^{-6} m
Particles	(In this context) the smallest pieces of which a substance is made; particles can be molecules, atoms or ions
Diffusion	The net movement of particles from a region of their higher concentration to a region of their lower concentration (i.e. down a concentration gradient), as a result of their random movement
Net movement	Overall or average movement
Concentration gradient	An imaginary 'slope' from a high concentration to a low concentration
Random movement	Movement in any direction, purely by chance; particles travel in straight lines and change direction when they collide with other particles
Kinetic energy	The energy of movement; the energy for diffusion and osmosis comes from the kinetic energy of the moving particles, not from respiration
Rate of diffusion	How fast particles diffuse; it is faster at a higher temperature, with a larger surface area, a steeper concentration gradient and a shorter diffusion distance
Diffusion distance	The distance particles have to travel, e.g. the thickness of a membrane; the shorter the distance, the faster the diffusion
Surface area to volume ratio	The surface area of an object divided by its volume; cutting a cube into smaller cubes increases this ratio, so substances diffuse in faster
Independent variable	The variable that you change in an experiment
Dependent variable	The variable that you measure as you collect your results
Standardised variable	A variable that you keep the same throughout an experiment so that it cannot affect the results (also called a control variable)
Solvent	The liquid in which a substance dissolves; in the solutions in and around cells, the solvent is water
Solute	A substance that is dissolved in a solvent, e.g. sugar dissolved in water
Partially permeable membrane	A membrane that lets some molecules through (e.g. small water molecules) but not others (e.g. much larger sugar molecules); cell membranes are partially permeable
Osmosis	The diffusion of water molecules through a partially permeable membrane

Osmosis (in terms of water potential)	The net movement of water molecules from a region of higher water potential (dilute solution) to a region of lower water potential (concentrated solution) through a partially permeable membrane
Water potential	A way of describing how much water a solution contains compared with dissolved solute; a dilute solution has a high water potential, a concentrated solution has a low water potential, and pure water has the highest water potential of all
High water potential	An area where there are a lot of water molecules, that is, a dilute solution
Low water potential	An area where there are not many water molecules, that is, a concentrated solution
Water potential gradient	A difference in water potential between two areas; water moves down it, from a high to a low water potential
Turgid	A description of a plant cell that is tight and firm
Turgor pressure	The pressure of the water pushing outwards on a plant cell wall
Flaccid	A description of a plant cell that has lost its turgor pressure and is soft
Plasmolysed	A description of a cell in which the cell membrane tears away from the cell wall
Plasmolysis	What happens when a plant cell loses so much water by osmosis that its cell membrane tears away from the cell wall; it can kill the cell
Wilting	The drooping of a plant that happens when its cells become flaccid and the plant loses its firmness
Active transport	The movement of molecules or ions through a cell membrane from a region of lower concentration to a region of higher concentration (i.e. against a concentration gradient) using energy from respiration
Carrier proteins (protein carriers)	Protein molecules in cell membranes that can use energy to change shape and move ions or molecules into or out of a cell
Root hair cell	A cell on the surface of a plant root that takes in water and mineral ions from the soil; it uses active transport to take in nitrate ions
Dialysis	A treatment for kidney failure in which unwanted substances diffuse out of the blood, through a very thin membrane, into dialysis fluid that then flows away
Indicator	A dye that changes colour depending on pH; cresol red is red-purple in alkaline conditions and turns yellow when acid is added
Anomalous result	A result that does not fit the pattern or trend shown by the other results
Biological molecules	Molecules that make up living cells, such as carbohydrates, fats and proteins; they are not often found in non-living things
Element	A substance made of only one kind of atom, e.g. carbon (C), hydrogen (H), oxygen (O) or nitrogen (N)
Compound	A substance made of two or more different elements chemically combined, e.g. glucose
Molecule	A particle made up of two or more atoms combined together, e.g. O_2 or $C_6H_{12}O_6$
Carbohydrates	Substances that include sugars, starch and cellulose; they contain carbon, hydrogen and oxygen, with about twice as many hydrogen atoms as carbon or oxygen atoms
Sugars	Carbohydrates that have relatively small molecules; they are soluble in water and they taste sweet
Glucose	A sugar that is used in respiration to release energy; formula $C_6H_{12}O_6$; the form in which carbohydrate is transported in the blood
Reducing sugar	A sugar, such as glucose, which turns Benedict's solution orange-red when they are heated together
Starch	A carbohydrate that is used as an energy store in plant cells; made of many glucose molecules in a long chain that coils into a spiral
Glycogen	A carbohydrate that is used as an energy store in animal cells; made of many glucose molecules in branched chains
Lipids	Substances containing carbon, hydrogen and oxygen; they are insoluble in water and are used as energy stores in organisms; another name for fats and oils
Fats	Lipids that are solid at room temperature
Oils	Lipids that are liquid at room temperature

Glycerol	One of the two kinds of smaller molecule that make up a fat; each fat molecule contains one glycerol molecule
Fatty acids	Smaller molecules that combine with glycerol to make fats and oils; each fat molecule contains three fatty acid molecules
Protein	A substance whose molecules are made of many amino acids linked together; each different protein has a different sequence of amino acids
Amino acids	Substances with molecules containing carbon, hydrogen, oxygen and nitrogen; there are 20 different amino acids found in organisms
Enzymes	Proteins that act as biological catalysts, speeding up chemical reactions in living organisms; all enzymes are proteins
Antibodies	Molecules secreted by white blood cells, which bind to pathogens and help to destroy them; antibodies are proteins
Pathogens	Microorganisms, such as bacteria, that cause disease
Haemoglobin	The red pigment that transports oxygen in mammalian blood; it is a protein
Keratin	The protein that forms hair (and fingernails)
Iodine solution	A solution of iodine in potassium iodide; it is orange-brown, and turns blue-black when mixed with starch
Benedict's solution	A blue liquid that turns orange-red when heated with a reducing sugar
Biuret reagent	A blue solution (also called biuret solution) that turns violet (purple) when mixed with proteins
Emulsion	A liquid containing two substances that do not fully mix; one of them forms tiny droplets dispersed throughout the other
Iodine test	The food test for starch: add iodine solution to the food; a blue-black colour means starch is present
Benedict's test	The food test for reducing sugars: heat the food with Benedict's solution in a water-bath; green, yellow or orange-red means reducing sugar is present
Biuret test	The food test for proteins: add biuret solution to the food and shake, with no heating; a violet (purple) colour means protein is present
Ethanol emulsion test	The food test for fats and oils: shake the food with ethanol, then pour the ethanol into water; a milky white emulsion means fat or oil is present
Water-bath	A container of water kept at a steady temperature, used to heat test-tubes gently and evenly, e.g. at about 80 °C for the Benedict's test
Catalyst	A substance that increases the rate of a chemical reaction and is not changed by the reaction
Biological catalyst	A catalyst made by living cells; enzymes are biological catalysts that speed up the reactions inside organisms
Substrate	The substance that an enzyme causes to react (for example, starch is the substrate of amylase)
Product	The new substance formed by a chemical reaction
Active site	The part of an enzyme molecule to which the substrate temporarily binds
Complementary	With a perfect mirror-image shape; the active site and its substrate have complementary shapes, so they fit together
Enzyme–substrate complex	The short-lived structure formed as the substrate binds temporarily to the active site of an enzyme
Specificity	Of enzymes, only able to act on a particular (specific) substrate
Lock and key model	The idea that a substrate fits an enzyme's active site like a key fits a lock; only a substrate with the complementary shape can fit
Carbohydrases	Enzymes that break down carbohydrates (for example, amylase, maltase, sucrase and lactase)
Protease	An enzyme that catalyses the breakdown of protein to amino acids
Lipases	Enzymes that break down lipids (fats and oils) into fatty acids and glycerol
Amylase	An enzyme that catalyses the breakdown of starch to maltose
Maltase	An enzyme that catalyses the breakdown of maltose to glucose
Sucrase	An enzyme that breaks down sucrose

Lactase	An enzyme that breaks down the milk sugar lactose into smaller sugar molecules that can be absorbed into the blood
Catalase	An enzyme that catalyses the breakdown of hydrogen peroxide to water and oxygen; it works inside all living cells
Hydrogen peroxide	A very dangerous substance made by many chemical reactions inside cells; catalase must break it down immediately
Lactose intolerance	Feeling ill (diarrhoea, abdominal pain, flatulence) after eating dairy foods, because the body does not make lactase and lactose is not broken down
Optimum	Best; for example, the optimum temperature of an enzyme is the temperature at which its activity is greatest
Optimum temperature	The temperature at which an enzyme works fastest; about 37 °C for enzymes from the human body
Optimum pH	The pH at which an enzyme works fastest; around pH 7 for most enzymes, but about pH 2 for a protease in the stomach
Denaturation	Damage to an enzyme's shape (including its active site) caused by high temperature or an unsuitable pH; the enzyme is denatured and can no longer bind its substrate or catalyse its reaction
Effective collision	A collision in which a substrate molecule enters an enzyme's active site, forms an enzyme–substrate complex and is converted to product
Range	The lowest to the highest value
Buffer solution	A liquid that has a known pH, and keeps that pH steady all the time, even if chemical reactions take place
Thymolphthalein	An indicator that is blue in alkaline conditions and loses its colour when the pH becomes less alkaline
Photosynthesis	The process by which plants synthesise carbohydrates from raw materials using energy from light
Organic substances	Carbon-containing biological molecules such as carbohydrates, fats and proteins; plants can make them but animals must take them in as food
Raw materials of photosynthesis	Carbon dioxide (from the air) and water (from the soil), which react to make glucose and oxygen
Chloroplast	The part of a plant cell where photosynthesis happens; contains chlorophyll (spread out on membranes) and the enzymes for photosynthesis
Sucrose	A sugar whose molecules are made of glucose and another similar molecule (fructose) linked together; the form in which plants transport carbohydrate
Nitrate ions	Mineral ions absorbed from the soil through the root hairs; they supply the nitrogen needed to make amino acids, proteins and chlorophyll
Magnesium ions	Mineral ions absorbed from the soil; magnesium is needed to make chlorophyll
Nectar	A sweet liquid secreted by many insect-pollinated flowers, to attract their pollinators
Pollen grains	Small structures which contain the male gametes of a flower
Pollinator	An animal (insect, bat or bird) that carries pollen grains from one flower to another
Xylem vessels	Tubes that carry water and mineral ions from the roots up to the leaves
Phloem tubes	Tubes that carry sucrose (and amino acids) from one part of a plant to another
Vascular bundles	Collections of xylem tubes and phloem vessels running side by side, which form the veins in a leaf
Lamina	The broad, flat part (blade) of a leaf
Midrib	The main vein running down the centre of a leaf
Leaf stalk	The stalk that joins the lamina to the rest of the plant; it contains vascular bundles
Dicotyledonous (dicot) plant	A flowering plant whose leaves usually have a branching network of veins
Transverse section (TS)	A thin slice cut across a structure, such as a leaf, so that its layers of cells can be seen under a microscope
Cuticle	A thin, transparent, waterproof covering made of a waxy substance, on the upper surface of a leaf
Upper epidermis	The upper layer of tightly packed, transparent cells that reduces water vapour loss and lets light through
Palisade mesophyll	A layer of tall, narrow cells with very large numbers of chloroplasts near the top of the leaf; the main site of photosynthesis
Spongy mesophyll	A layer of photosynthesising cells with many air spaces between them, allowing gases to diffuse through the leaf
Air spaces	Gaps between spongy mesophyll cells through which carbon dioxide, oxygen and water vapour diffuse
Lower epidermis	The bottom layer of cells of a leaf; it contains the stomata and usually has no cuticle
Stoma (plural stomata)	An opening in the lower epidermis that lets carbon dioxide and oxygen diffuse in and out, and water vapour diffuse out
Guard cells	A pair of cells (containing chloroplasts) surrounding each stoma; they change shape to open and close it
Variegated leaf	A leaf that has some white areas (no chlorophyll) and some green areas (with chlorophyll)
Destarching	Leaving a plant in the dark for long enough for it to use up its starch stores
Control	A standard sample that you use as a comparison, to find the effect of changing a variable
Potassium hydroxide solution	A solution that absorbs carbon dioxide; used to remove carbon dioxide from the air around a leaf
Sodium hydrogencarbonate solution	A solution that provides carbon dioxide for aquatic plants; a higher concentration supplies more carbon dioxide
Rate of photosynthesis	How much photosynthesis happens in unit time, e.g. the number of oxygen bubbles released by an aquatic plant in one minute
Limiting factor	The factor in shortest supply (light intensity, carbon dioxide concentration or temperature) that stops the rate of photosynthesis going any higher
Heat shield	A transparent barrier between a lamp and a plant that lets light through but stops the lamp's heat changing the temperature
Glasshouse	A glass building in which temperature, light and carbon dioxide can be controlled so that crops photosynthesise quickly and give a high yield
Biofuels	Fuels made from plants grown for the purpose, which can be burnt to produce electricity or move vehicles

Key Equations

- cells → tissues → organs → organ systems → organism
- Aerobic respiration (in mitochondria, details in Chapter B9): glucose + oxygen → carbon dioxide + water, releasing energy
- Diffusion: higher concentration → lower concentration (down a concentration gradient) by random movement; the energy is the particles' kinetic energy, not respiration
- Osmosis: higher water potential (dilute solution) → lower water potential (concentrated solution), through a partially permeable membrane
- Active transport: lower concentration → higher concentration (against a concentration gradient), through carrier proteins in the cell membrane, using energy from respiration
- many glucose molecules → starch (the energy store in plants)
- many glucose molecules → glycogen (the energy store in animals)
- Iodine test: orange-brown → blue-black means starch is present.
- Benedict's test: blue → green → yellow → orange-red as the amount of reducing sugar increases.
- 1 glycerol + 3 fatty acids → 1 fat (lipid) molecule
- Ethanol emulsion test: clear → milky white emulsion means fat or oil is present.
- Many amino acids (20 kinds) joined in a specific sequence → one protein molecule
- Biuret test: blue → violet (purple) means protein is present.
- Many glucose molecules → starch (energy store in plants) or glycogen (energy store in animals)
- 1 glycerol + 3 fatty acids → 1 fat or oil (lipid) molecule
- Many amino acids (20 kinds) in a specific sequence → a protein; a different sequence gives a different protein
- Iodine test for starch (no heating): orange-brown → blue-black
- Benedict's test for reducing sugars (heat in a water-bath at about 80 °C): blue → green → yellow → orange-red as the concentration of reducing sugar increases
- Biuret test for proteins (no heating): blue → violet (purple)
- Ethanol emulsion test for fats and oils: shake the food with ethanol, pour the ethanol into water → milky white emulsion
- starch → maltose (catalysed by amylase)
- protein → amino acids (catalysed by protease)
- hydrogen peroxide → water + oxygen (catalysed by catalase)
- enzyme + substrate → enzyme–substrate complex → enzyme + product(s)

- lipid → fatty acids + glycerol (catalysed by lipase)
- Maltase: maltose → glucose
- Lipase: lipid (fats and oils) → fatty acids + glycerol
- Lactase: lactose → smaller sugars (glucose + galactose), which are absorbed into the blood
- Word equation: carbon dioxide + water → glucose + oxygen (with "sunlight" written above the arrow and "chlorophyll" below it)
- Balanced chemical equation: $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$
- Glucose + nitrogen (from nitrate ions) → amino acids → proteins
- Glucose + nitrogen (from nitrate ions) + magnesium (from magnesium ions) → chlorophyll
- Carbon dioxide route: atmosphere → stomata → air spaces → palisade cells (by diffusion)
- Water route: soil → roots → xylem vessels in the veins → leaf cells (by osmosis)

Exam Focus

1. Learn MRS GREN with definitions: movement, respiration, sensitivity, growth, reproduction, excretion and nutrition. Questions often ask which feature is NOT a characteristic of all organisms: photosynthesis, breathing, a blood system and a nervous system are not.
2. Respiration is not breathing: respiration is a set of chemical reactions in every living cell that releases energy from glucose; breathing is moving air in and out of the lungs, which plants and bacteria do not do.
3. Growth must be permanent: define growth as a permanent increase in size and dry mass; the word permanent earns the mark.
4. Know which structures are in which cells: animal and plant cells both have a cell membrane, cytoplasm, a nucleus, ribosomes and mitochondria; only plant cells have a cellulose cell wall, a large permanent vacuole with cell sap and (often) chloroplasts and starch grains.
5. Bacteria have no nucleus: their DNA is a single circle in the cytoplasm, often with plasmids; their cell wall is not made of cellulose; they have no mitochondria or chloroplasts.
6. Partially permeable versus fully permeable: the cell membrane is partially permeable and controls what enters and leaves the cell; the cell wall is fully permeable, supports the cell and stops it bursting, but does not control what enters.
7. Mitochondria release energy: write that mitochondria release energy from glucose by aerobic respiration; never write that they make or produce energy.
8. Ribosomes make proteins; DNA decides which proteins: the DNA in the nucleus carries the instructions, and the ribosomes link amino acids together to build the protein.
9. No chloroplasts does not mean animal cell: many plant cells, such as root hair cells, have no chloroplasts; the cell wall is the reliable sign of a plant cell.
10. Cell, tissue, organ, organ system, organism: a tissue is a group of similar cells; an organ is made of different tissues; be ready to give an example at each level (stomach lining cell, stomach lining tissue, stomach, digestive system, human).
11. Show the magnification equation and substitution: write magnification = image size ÷ actual size, convert to the same units, substitute, then give the answer with a × sign and no units; each step can earn a mark.
12. Divide to find the actual size: actual size = image size ÷ magnification; multiplying instead of dividing is the most common error in multiple-choice questions.
13. Convert mm and µm correctly: multiply by 1000 to go from mm to µm, divide by 1000 to go from µm to mm, and check that your final size is sensible for the structure.
14. Measure carefully and record the measurement: measure in mm with a ruler along the length asked for, and write the measurement in your working.
15. Use the marks as a guide: a [6] question needs six separate creditworthy points, and a [2] definition usually needs two ideas.
16. Learn the full definition of diffusion: include "net movement", "from a region of higher concentration to a region of lower concentration", "down a concentration gradient" and "as a result of random movement"; each phrase can earn credit.
17. Diffusion and osmosis are passive: their energy comes from the kinetic energy of the particles; never say they use energy from respiration, because only active transport does.
18. Osmosis is only about water: it is the diffusion of water molecules through a partially permeable membrane; the solute does not move in osmosis.
19. Use water potential correctly: a dilute solution has a high water potential, a concentrated solution has a low water potential, and water moves from high to low water potential.
20. Explain why plant cells do not burst: the strong cell wall resists the swelling contents and presses back; use the words turgid and turgor pressure, and say animal cells burst because they have no cell wall.
21. Do not mix up flaccid and plasmolysed: a flaccid cell has lost its turgor pressure and is soft; a plasmolysed cell has lost so much water that its cell membrane has torn away from the cell wall.
22. The cell wall is fully permeable: it is the cell membrane, not the cell wall, that is partially permeable, so the space between the wall and the membrane of a plasmolysed cell fills with the outside solution.
23. Active transport answers need three ideas: movement against a concentration gradient (low to high), carrier proteins in the membrane that change shape, and energy from respiration in mitochondria.
24. Quote data from graphs and bar charts: give values with units and compare them (for example "potassium is 8 times more concentrated in the cells"), then use the data to decide which process is happening.
25. Identify anomalous results properly: pick the point that is furthest from the line of best fit and does not fit the trend, and say why you chose it.
26. Plan experiments using variables: name the independent, dependent and standardised variables, repeat readings and calculate a mean, and put units only in the column headings.
27. Show working in surface area calculations: use cm² for area and cm³ for volume; cutting a cube into smaller cubes increases the total surface area but not the total volume.
28. Common mistake: saying particles want to move: particles move randomly; the net movement happens because more particles move away from the high concentration than move into it.
29. Know the elements in each molecule: carbohydrates and fats contain only C, H and O; proteins contain C, H, O and N. Nitrogen is the element found in all proteins but not in carbohydrates or fats.
30. Link each large molecule to its smaller building blocks: starch and glycogen are made from glucose; fats and oils from fatty acids and glycerol; proteins from amino acids. A common mistake is to say that fats are made of amino acids, or that starch is made of "sugar" without naming glucose.
31. Starch is found in plants, glycogen in animals: glucose is found in both. Starch is the energy store in plant cells and glycogen is the energy store in animal cells.
32. Only the Benedict's test needs heating: always write "heat in a water-bath (about 80 °C)". Never heat the biuret test; this is the error in Practice Question 4a(i).
33. Always give the colour before and after: iodine orange-brown → blue-black; Benedict's blue → orange-red; biuret blue → violet; ethanol test → milky white emulsion. Write the colour you actually see (e.g. "stayed blue"), never just "no change" or "positive".
34. Do not mix up the colours: iodine solution starts orange-brown, not blue. Benedict's and biuret both start blue, but Benedict's ends orange-red and biuret ends violet. Swapping the start and end colours, as in Practice Question 4a(ii), loses the mark.
35. Describe the ethanol emulsion test in the right order: shake the food with ethanol first, then pour the ethanol (not the food) into water; a milky white emulsion means fat. Use very clean tubes.
36. Read results tables carefully: "contains only fat" means positive for the ethanol emulsion test but negative for every other test (iodine stays brown, Benedict's stays blue, biuret stays blue).
37. Benedict's tests for reducing sugars, not starch: the colour reached (green, yellow or orange-red) shows how much reducing sugar is present.
38. Use the command words: identify (name, select or recognise), give (produce an answer from a given source or from recall), describe (state the points of a topic or give its main features). For "describe how you could test", give the reagent, what you do and the result.
39. Learn the definition of a catalyst exactly: "a substance that increases the rate of a chemical reaction and is not changed by the reaction". You need both parts: it speeds up the reaction and it is unchanged.
40. Enzymes are proteins: they are made of amino acids and act as biological catalysts in all metabolic reactions. Without them, reactions would be too slow to sustain life.
41. Use the word complementary: the active site has a shape that is complementary to the substrate. Do not say the active site is "the same shape" as the substrate.
42. Name the enzyme–substrate complex: in Supplement answers, say that the substrate binds to the active site to form an enzyme–substrate complex, the products are released, and the enzyme is unchanged.
43. Explain specificity using shape: only a substrate with a shape complementary to the active site can fit and bind, so each enzyme acts on only one type of substrate.
44. Enzymes are denatured, not killed: enzymes are molecules, not living things, so never write that they "die" or are "killed".
45. Low temperature does not denature enzymes: at low temperatures enzymes are slow because molecules have little kinetic energy; they work again when warmed. Denaturation happens above the optimum temperature or at a pH far from the optimum.
46. Use the key words for temperature: kinetic energy, frequency of effective collisions, enzyme–substrate complex, shape of the active site, complementary, denatured.
47. Explain pH effects in terms of shape: pH changes the shape of the enzyme and its active site. Kinetic energy and collisions are the explanation for temperature, not for pH.

48. Describe graphs with numbers and units: quote the optimum (for example "about 37 °C") and the values where activity starts and stops. "Describe" means say what happens; "explain" means give the reasons.
49. Time and rate are opposites: a shorter time means a faster rate (rate = 1 ÷ time). In the catalase, lipase and amylase experiments, the shortest time is at the optimum.
50. Know the iodine test colours: blue-black means starch is present; brown means no starch. Do not say that iodine "turns clear".
51. Spell enzyme names carefully: maltose (a sugar) and maltase (an enzyme) look almost the same; lipid is the substrate and lipase is the enzyme.
52. Make controlled variables specific: write "temperature", "volume of amylase solution" or "concentration of starch solution", not just "the amount".
53. Different enzymes have different optimums: most enzymes about pH 7 but the stomach protease about pH 2; human enzymes about 37 °C but some hot-spring bacteria up to 80 °C.
54. Learn both photosynthesis equations exactly: Word equation: carbon dioxide + water → glucose + oxygen. Balanced equation: $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$. Write light and chlorophyll on the arrow, never as reactants, and check that the atoms balance.
55. Chlorophyll transfers energy, it does not make it: Say that chlorophyll absorbs (traps) light energy and transfers it into chemical energy in glucose. Never write that plants "make energy".
56. Know the starch test and the reason for every step: Boil in water (breaks cell membranes so iodine can reach the starch), turn off the flame, warm alcohol in a water bath (removes chlorophyll so the colour can be seen), soften in hot water, add iodine solution: blue-black means starch, orange-brown means none.
57. Explain destarching: The plant is kept in the dark for about two days so it uses up its starch; any starch found later must have been made during the experiment.
58. Always identify the control: Green parts of a variegated leaf (chlorophyll), the uncovered part of the leaf (light), the leaf in the flask of distilled water (carbon dioxide). The control is identical except for the factor being tested.
59. Name the leaf tissues in order and link each to its function: Cuticle, upper epidermis, palisade mesophyll, spongy mesophyll with air spaces, vascular bundle (xylem and phloem), lower epidermis with stomata and guard cells.
60. Know which cells contain chloroplasts: Palisade mesophyll cells have the most, spongy mesophyll cells have many, guard cells have a few; other epidermis cells and xylem have none. Use this to interpret data such as the bar chart in Practice Question 4.
61. Get the routes of the raw materials right: Carbon dioxide diffuses in through the stomata and air spaces; water travels up the xylem and then moves into cells by osmosis. Carbon dioxide does not enter through the roots.
62. Give a reason with each use of glucose: Starch (insoluble store, no effect on osmosis), sucrose (transport in phloem), cellulose (cell walls), amino acids and proteins (growth, need nitrate), chlorophyll (needs nitrate and magnesium), nectar (attracts pollinators), respiration (releases energy).
63. Do not mix up nitrate and magnesium: Nitrate ions supply nitrogen for amino acids and proteins (deficiency: weak growth, yellow leaves); magnesium ions are needed for chlorophyll (deficiency: yellowing, especially between the veins).
64. Plan rate experiments with clear variables: State the independent variable (light intensity via lamp distance, temperature, or carbon dioxide concentration), the dependent variable (bubbles or gas volume per minute), the controlled variables, the heat shield, and repeats with means.
65. Draw graphs that earn every mark: Independent variable on the x-axis, both axes labelled with units, even scales filling at least half the grid, small neat crosses, ruled straight lines from point to point, nothing extended beyond the data.
66. Describe, then explain, data: Describe the trend with numbers from the table or graph (for example "increased from 5 mm to 17 mm"), then explain it using raw materials, energy from light, enzymes or limiting factors.
67. Avoid common mistakes: Photosynthesis does not "change carbon dioxide into oxygen" (the carbon goes into glucose); plants do not get their food from the soil (only water and mineral ions); bubble counts are not a perfect measure of oxygen because bubbles vary in size.

Key Facts

- Plant cells have very large, permanent vacuoles, which contain a solution of sugars and other substances called cell sap. When the vacuole is full, it presses outwards on the rest of the cell and helps it to keep its shape.
- Large vacuole: a big, clear space in the middle of a plant cell that pushes the cytoplasm and nucleus out to the edges.
- Ciliated cell: has tiny hair-like cilia on its surface that beat to sweep mucus (with trapped dust and bacteria) up the trachea and bronchi.
- Mixing units: dividing a length in mm by a length in µm gives an answer that is 1000 times too big or too small. Convert first.
- Movement: plants cannot move their whole body from place to place, but they slowly move parts of themselves; for example, many flowers open during the day and close at night.
- Net movement means overall or average movement. Some molecules go from the egg into the room and a few go from the room back into the egg, but overall most of them go from the egg into the room.
- Plasmolysed: when the cell membrane tears away from the cell wall, the cell is said to be plasmolysed. Plasmolysis can kill a plant cell, because the cell membrane is damaged as it tears away from the cell wall.
- Element — a substance made of only one kind of atom, such as carbon (C), hydrogen (H), oxygen (O) or nitrogen (N)
- Glycogen (in animals): glucose molecules link together to form glycogen, which is used as an energy store in animal cells. The liver helps to keep the concentration of glucose in the blood constant (you can read about this in Chapter B9)
- Heat is essential: the mixture must be heated in a water-bath (about 80 °C), otherwise there is no colour change
- Cell membranes: fats are important for making cell membranes
- Positive result: it changes to violet (purple) when it is in contact with proteins; a pale lilac colour means only a little protein
- Spelling warning: you have to be careful to spell the name of an enzyme and its substrate correctly and to write clearly. Maltose and maltase can look almost the same if you do not write carefully. A useful rule: names ending in -ose are sugars (maltose, sucrose, lactose), and names ending in -ase are enzymes (maltase, sucrase, lactase).
- Few data points: test more pH values and do more repeats, then check that the repeats are close together.
- Ideas for the demonstration: show the froth forming on raw potato; explain the active site with a simple model (for example, two shapes that fit together like a lock and key); compare raw potato with boiled potato, where the catalase has been denatured and no froth forms; show that the gas relights a glowing splint, proving it is oxygen.
- At the optimum (shortest time): the rate is greatest, because there are the most effective collisions while the active site still has its correct shape.
- H_2O (water): one molecule contains 2 atoms of hydrogen (H) and 1 atom of oxygen
- Veins — the vascular bundles branching through the lamina. The tubes in the vascular bundles carry substances to and from the leaf (Chapter B7).
- Variables: Nothing is varied — this is a standard test that you will use in Experimental skills B5.02, B5.03 and B5.04.
- Respiration [1] — glucose is broken down to release energy, for example for active transport of mineral ions into root hairs, or for building proteins [1].
- Bacterial cells: they have cytoplasm, a cell membrane, a cell wall and ribosomes, but no nucleus. Their DNA is circular, and they may have extra, small circles of DNA called plasmids.
- Define: give a precise meaning.
- State: express in clear terms.
- As temperature increases, the kinetic energy of enzyme and substrate molecules increases, and they collide more frequently. This increases the rate of activity of the enzyme. However, above the optimum temperature, the enzyme begins to lose its shape, until it can no longer combine with the substrate. The shape of the active site is no longer complementary to the shape of the substrate.
- Diagrams of enzyme action: label the substrate, active site, enzyme–substrate complex and products, and say that the enzyme is unchanged at the end.

Quick Q&A

Question	Answer
How many micrometres are there in 1 mm?	There are 1000 µm in 1 mm. To convert millimetres to micrometres, multiply by 1000.
The diagram below shows a cell from a plant leaf. Identify which labelled part contains DNA. (1 mark) A) the outer double line that forms the boundar	B — the small oval structure is the nucleus, which contains DNA in the form of chromosomes. (A is the cell wall, C is the cytoplasm and D is the large
A diagram of a flower is 8 cm across. The magnification of the diagram is ×4. Identify the actual width of the flower. (1 mark) A) 3.2 mm B) 20 mm C)	B — 20 mm. Actual size = size of image ÷ magnification = 8 cm ÷ 4 = 2 cm, and 2 cm × 10 = 20 mm. D (32 cm) comes from multiplying instead of dividing,
These questions are about the diagram in Figure B2.04, which shows a dilute sugar solution (left) separated from a concentrated sugar solution (right)	The solvent is water. The solute is sugar.
Copy and complete this sentence: When an animal cell is placed in pure water, water …… the cell by …… through the partially permeable cell ……	When an animal cell is placed in pure water, water enters (moves into) the cell by osmosis through the partially permeable cell membrane.
You read about the enzyme lactase at the start of this chapter. a) What type of	a) Lactase is a carbohydrase, because it breaks down a carbohydrate (the sugar lactose) into smaller sugar

enzyme is lactase? b) Name the substrate of lactase.	molecules. b) Lactose, the sugar found in
The figure below shows some stages in an enzyme-controlled reaction. (4 marks) (The figure shows three stages, from top to bottom. Stage 1: an enzyme	P = substrate [1] Q = active site [1] R = enzyme–substrate complex [1] S = products (the product molecules) [1]

Animals do not make or store starch. What is the substance that animals store, which is made of chains of glucose molecules?	Glycogen (stored mainly in the liver and muscles).